POWER PULSE

Measurement of Power Important Formulae

Notes by Balaji Sir

1 Wattmeter

1. $$P = V\,I\cos\phi, \qquad \text{Multiplying factor} = \frac{V_{range}\,I_{range}\cos\phi}{\text{full-scale divisions}}$$

reading = V I cos φ × (divisions/FSD)

2. $$\text{Reactive power by one wattmeter (balanced)}: \quad Q = \sqrt{3}\,W$$

the wattmeter has its pressure coil across the other two lines

2 Two-wattmeter method (balanced load)

3. $$W_1 = V_L I_L \cos(30^\circ - \phi), \qquad W_2 = V_L I_L \cos(30^\circ + \phi)$$

4. $$P = W_1 + W_2, \qquad Q = \sqrt{3}\,(W_1 - W_2)$$

5. $$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}, \qquad \text{pf} = \cos\phi$$

φ is the load phase angle

6. $\text{pf} = 1:\ W_1 = W_2;\ \ \text{pf} = 0.5:\ W_2 = 0;\ \ \text{pf} < 0.5:\ W_2 < 0;\ \ \text{pf} = 0:\ W_1 = -W_2$

W_2 is the lower reading

3 Three-wattmeter method

7. $$P = W_1 + W_2 + W_3 \ \ (\text{3-phase, 4-wire})$$

4 Pressure-coil inductance error

8. $$\tan\beta = \frac{\omega L_p}{R_p}, \qquad \frac{P_{indicated}}{P_{true}} = \frac{\cos\beta\,\cos(\phi - \beta)}{\cos\phi} \approx 1 + \tan\phi\tan\beta$$

error is larger at a low power factor; a compensating capacitor across part of the series resistance reduces it